exponentially at the rate of 0.6% per year. If the growth continues at this rate, how many tons of gold will be produced from 2000 to 2013? [$e^{0.078} = 1.0811$)

Solution:

Annual consumption at time $t = 0$ (In the year 2000) : $P_0 = 2547$ metric ton.

Total production of Gold from 2000 to 2013 $= \int_0^{13} 2547\, e^{0.006t}\, dt$

$$= \frac{2547}{0.006}\left[e^{0.006t}\right]_0^{13}$$

$$= 424500\left(e^{0.078} - 1\right)$$

= 34,426.95 metric tons approximately.

3.2.3 The demand functions from elasticity of demand

Elasticity of the function $y = f(x)$ at a point x is defined as the limiting case of ratio of the relative change in y to the relative change in x.

$$\therefore \quad \eta = \frac{E_y}{E_x} = \lim_{\Delta_x \to 0} \frac{\frac{\Delta y}{y}}{\frac{\Delta x}{x}} = \frac{\frac{dy}{y}}{\frac{dx}{x}}$$

$$\Rightarrow \quad \eta = \frac{x}{y}\cdot\frac{dy}{dx}$$

Elasticity of demand $\eta_d = \frac{-p}{x}\frac{dx}{dp}$

$$\frac{-dp}{p} = \frac{dx}{x}\cdot\frac{1}{\eta_d}$$

Integrating both sides w.r. to x

$$-\int\frac{dp}{p} = \frac{1}{\eta_d}\int\frac{dx}{x}$$

Equation yields the demand function 'p' as a function of x.

The revenue function can be found out by using integration.

Example 3.25

When the Elasticity function is $\frac{x}{x-2}$.

Find the function when $x = 6$ and $y = 16$.

Solution:

$$\frac{E_y}{E_x} = \frac{x}{x-2}$$

$$\frac{x}{y}\frac{dy}{dx} = \frac{x}{x-2}$$

$$\frac{dy}{y} = \frac{x}{x-2}\cdot\frac{dx}{x}$$

$$\int\frac{dy}{y} = \int\frac{dx}{x-2}$$

$$\log y = \log(x-2) + \log k$$

$$y = k(x-2)$$

when $x = 6, y = 16 \Rightarrow 16 = k(6-2)$

$$k = 4$$

$$y = 4\,(x-2)$$

Example 3.26

The elasticity of demand with respect to price p for a commodity is $\eta_d = \frac{p + 2p^2}{100 - p - p^2}$.

Find demand function where price is ₹5 and the demand is 70.

Solution:

$$\eta_d = \frac{p + 2p^2}{100 - p - p^2}$$

$$\frac{-p}{x}\frac{dx}{dp} = \frac{p(2p+1)}{100 - p - p^2}$$

$$-\frac{dx}{x} = \frac{-(2p+1)}{p^2 + p - 100}dp$$

$$\int\frac{dx}{x} = \int\frac{2p+1}{p^2 + p - 100}dp$$

$$\log x = \log(p^2 + p = 100) + \log k$$

$$\therefore x = k(p^2 + p - 100)$$

When $x = 70, \quad p = 5,$

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